

Abu Jafar Saifullah

☎ +880 1717 743519 | @jafarmahin107@gmail.com | 🌐 LinkedIn | 🐙 GitHub | 📍 Dhaka, Bangladesh

RESEARCH INTERESTS

AI for Software Engineering, Automated Program Repair, Software Testing, Dynamic Analysis, Human-Centered Software Engineering (Web Accessibility)

EDUCATION

University of Dhaka

M.Sc. in Software Engineering (Final Semester); GPA: 3.92/4.00

B.Sc. in Software Engineering; GPA: 3.89/4.00

Dhaka, Bangladesh

Apr 2025 – Aug 2026

Jan 2019 – Dec 2023

Notre Dame College

Higher Secondary Certificate; GPA: 5.0/5.0

Dhaka, Bangladesh

Jun 2014 – Jun 2016

PUBLICATIONS

Abu Jafar Saifullah, Tasmia Zerin, Zerina Begum, Kazi Sakib. “NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation.” In *Proceedings of the 21st International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE 2026)*. Benidorm, Spain. Presented by author, May 2026. [\[Code\]](#)

Key results: 100% precision and recall on a 22-page controlled dataset (32 SC-level assertions); 2,947 violations detected across 26 production websites; 90% precision confirmed via independent manual validation (PABAK = 0.80). First tool to cover all five WCAG 2.4 focus-behavior success criteria.

RESEARCH EXPERIENCE

Institute of Information Technology, University of Dhaka

Graduate Researcher, supervised by Dr. Kazi Sakib & Zerina Begum

Dhaka, Bangladesh

Apr 2025 – Present

- Developing **NavA11y**, a dynamic analysis tool for automated WCAG 2.4 accessibility evaluation using Playwright-driven browser automation. Research forms the core of M.Sc. thesis.
- Designed a two-phase analysis architecture: Page-Level Analysis (keyboard traversal, Kendall τ divergence scoring) and Element-Level Analysis (focus-state inspection, 7×7 grid obscuration probing).
- Constructed and released a 22-page labeled focus-behavior dataset (*Dd*) covering all five WCAG 2.4 focus-behavior SCs, including pass, boundary, and obscuration cases.
- Conducted manual evaluation with two independent expert raters across 6 production websites; computed inter-rater agreement (PABAK = 0.80).

SEMERU Lab, William & Mary

Research Intern, supervised by Dr. Denys Poshyvanyk

Remote

Aug 2023 – Dec 2023

- Built a programming language-specific tokenizer to improve NLP model predictions for software engineering tasks, contributing to the lab’s research on intelligent software tools.

INDUSTRY EXPERIENCE

Optimizely, Content Marketing Platform

Software Engineer II (promoted from SE I, Apr 2026)

Dhaka, Bangladesh

Apr 2024 – Present, Full-time

- Contributed to the platform’s migration to **Axiom** ([@optiaxiom/react](#), public design system). Landed **26 merged PRs** in the open-source repo — V1→V2 component conversions plus net-new components (Aside, SubNav, Range) — and 10+ adapter components in the product. Wrote the Migration Guide and Icon Contribution Guide used by other product teams.
- Worked on end-to-end integration of an internal AI SDK. Migrated the backing service from Express to Hono, set up the AWS CodeArtifact publishing pipeline, and instrumented AI-context events across 10+ product surfaces.
- Implemented backend block-context extraction for LLM prompts, frontend block-form interaction, and AI resource rendering for the platform’s AI document UI (Adaptive-Cards-style pattern).
- Implemented **Dark Mode** across 15+ surfaces (CSS-variable refactor) and contributed to **Spotlight** global search (built during, and won, the **Optimizely Global Hackathon 2025** as part of a team); reviewed **300+ pull requests** across design-system and AI-tracking work.

Orion Informatics Ltd.

Software Engineer

Dhaka, Bangladesh

Jul 2023 – Mar 2024, Full-time

- Built data visualisation and reporting features for an Attendance Management System; developed a two-service ASP.NET Core webhook relay for TradeStation alerts deployed on Azure.

RESEARCH & ACADEMIC PROJECTS

NavA11y | *Node.js, Playwright, Chromium, GitHub Actions* | [Code](#) | *ENASE 2026*

- Dynamic WCAG 2.4 accessibility testing tool, the first to cover all five focus-behavior SCs. Two-phase architecture: Page-Level Analysis (Kendall τ divergence scoring of tab traversal) and Element-Level Analysis (7×7 grid obscuration probing, CSS focus-state inspection). 100% precision/recall on 22-page labeled benchmark; 2,947 violations on 26 production sites; 90% precision confirmed by independent expert raters (PABAK = 0.80). Dataset released for community benchmarking.

RepairA11y | *Node.js, Playwright, Vitest* | *M.Sc. Thesis, In Progress, 2026*

- LLM-driven accessibility repair pipeline evaluating whether runtime-evidence prompting (live DOM + NavA11y violation report) outperforms static-evidence prompting (HTML/screenshots) for automated WCAG 2.4 patch generation. Addresses four research questions covering effectiveness, evidence ablation, and regression safety.

Clippy | *Angular, Node.js, AWS S3* | [GitHub](#) | *ICSE SRC 2023, Top 6*

- Smart PDF reader with knowledge graph extraction, citation network visualization, and paper recommendation; submitted to ICSE 2023 Student Research Competition. 392 commits, 5 GitHub stars.

BPE Tokenizer | *Python, Flask, HuggingFace* | [PyPI](#)

- Byte Pair Encoding tokenizer implemented from scratch and published as `python-bpe-tokenizer` on PyPI; includes a Flask web interface for training on custom corpora. Built as direct precursor to SEMERU Lab tokenizer research.

ML Bug Mining (SZZ) | *Python, GitPython, pandas, scikit-learn* | *M.Sc. Coursework*

- Mined ML-related bugs across 9 open-source repositories using the SZZ algorithm; identified bug-fixing and bug-introducing commits via keyword heuristics and git blame, filtered for ML-library-touching code.

Blueberry Auto | *React, Hono, Supabase, Vercel* | [Live](#) | *Lead Developer, 2025 – Present*

- Vehicle marketplace platform built as a pnpm monorepo (frontend, Hono API, Supabase backend, crawler); leading end-to-end build as the contracted developer (**150+ merged PRs**).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Web & Frontend: React 18/19, Next.js, Angular, Node.js, Hono, Express, FastAPI, Spring Boot 3, ASP.NET Core

Testing & Automation: Playwright, axe-core, Vitest, Karma, Mocha, Chai, Sinon, Selenium

ML / NLP: TensorFlow, HuggingFace Transformers, BPE Tokenization (own impl.), TextRank, TF-IDF, scikit-learn, pandas

Research Methods: Dynamic Program Analysis, Mining Software Repositories (SZZ), Controlled User Studies, Statistical Evaluation (Kendall τ , Cohen's κ , PABAK), Inter-rater Agreement

DevOps / Cloud: AWS CodeArtifact, AWS S3, Azure, Cloudflare Workers, Docker, NGINX, GitHub Actions

Databases: MongoDB, PostgreSQL (Supabase), SQL Server, Firebase, Redis

AWARDS & ACHIEVEMENTS

ICT Division Fellowship, Government of Bangladesh: research funding for NavA11y / M.Sc. thesis (Grant No. 56.00.0000.052.33.002.24-61, Apr 2026)

Winner, Optimizely Global Hackathon 2025 (company-wide international hackathon; team built Spotlight, shipped to CMP)

Top 6, ICSE SCORE 2023 (international undergraduate SE competition)

Champion, Refactoring Competition (2023) **Best UI/UX,** Hackathon (2023)

Champion, Project Showcasing (2022) **Runners-up,** Project Showcasing (2021)